Math 125

Opportunity B

Name: ____________________

USC ID: ____________________

Instructions:

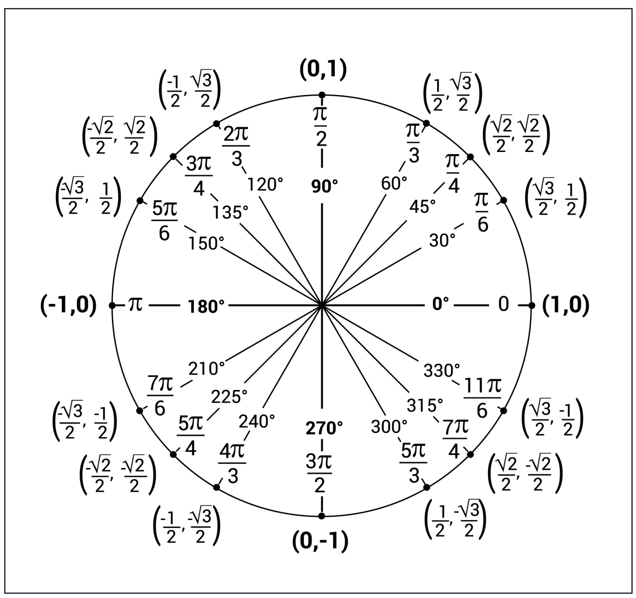
- Make sure to clearly show each step of your solution for full credit - correct answers without any work will not receive credit. Some questions may remind you to show your work explicitly, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Write all of your work and answers on this document. If you need extra paper, please ask the instructor. If a box is provided for your final answer, please write your final answer in the box.
- Indicate your affirmation of the following statement by signing below.

I affirm that I did not give or receive any unauthorized assistance to solve the problems below. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: ____

Formulas

- $L(x) = f'(a)(x-a) + f(a)$
- MVT: $\dfrac{f(b)-f(a)}{b-a} = f'(c)$
- Basic derivatives
  - $\dfrac{d}{dx}(x^n) = nx^{n-1}$
  - $\dfrac{d}{dx}(e^x) = e^x$
  - $\dfrac{d}{dx}(\ln(x)) = \dfrac{1}{x}$
  - $\dfrac{d}{dx}(\sin x) = \cos x$
  - $\dfrac{d}{dx}(\cos x) = -\sin x$
  - $\dfrac{d}{dx}(\tan x) = \sec^2 x$
  - $\dfrac{d}{dx}(\sec x) = \sec x \cdot \tan x$
  - $\dfrac{d}{dx}(\cot x) = -\csc^2 x$
  - $\dfrac{d}{dx}(\csc x) = -\csc x \cdot \cot x$
- Derivative rules
  - $\dfrac{d}{dx}(f(x)\cdot g(x)) = f'(x)\cdot g(x) + g'(x)\cdot f(x)$
  - $\dfrac{d}{dx}\left(\dfrac{f(x)}{g(x)}\right) = \dfrac{g(x)\cdot f'(x) - f(x)\cdot g'(x)}{[g(x)]^2}$
  - $\dfrac{d}{dx}(f(g(x))) = f'(g(x))\cdot g'(x)$



B1 - Extrema of functions (inspired by Brightspace problem 4 and past final worksheet problem 4)

Consider the following function along with its second derivative.

$$f(x) = 9x^{\frac{2}{3}}e^{\frac{x}{3}} \qquad\qquad f''(x) = \frac{e^{\frac{x}{3}}(x^2+4x-2)}{x^{\frac{4}{3}}}$$

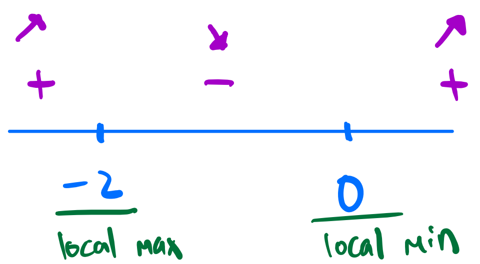
You don't need to check that the second derivative is correct, and can use it to answer any of the following questions.

(a) Show that $f'(x) = \dfrac{3e^{\frac{x}{3}}(x+2)}{x^{\frac{1}{3}}}$. Then find and classify all local extrema of $f(x)$.

$$f'(x) = 9x^{\frac{2}{3}}\cdot\tfrac{1}{3}e^{\frac{x}{3}} + 9\cdot\tfrac{2}{3}x^{-\frac{1}{3}}e^{\frac{x}{3}} = 3e^{\frac{x}{3}}\left(x^{\frac{2}{3}} + \frac{2}{x^{\frac{1}{3}}}\right) = 3e^{\frac{x}{3}}\left(\frac{x+2}{x^{\frac{1}{3}}}\right) = \frac{3e^{\frac{x}{3}}(x+2)}{x^{\frac{1}{3}}}$$

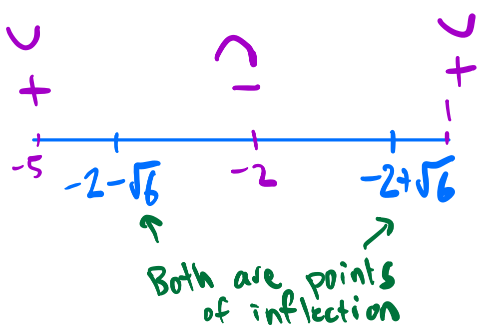
$x=2$

$x=0$ (DNE)



(b) Find all points of inflection of $f(x)$. Be sure to fully justify your answer.

$$x^2+4x-2=0 \rightarrow \quad x = \frac{-4\pm\sqrt{16+8}}{2} = \frac{-4\pm\sqrt{24}}{2} = \frac{-4\pm2\sqrt{6}}{2} = -2\pm\sqrt{6}$$



$f''(-5) = +$

$f''(-2) = -$

$f''(1) = +$

(c) Find the values of the absolute extrema of $f(x)$ on the domain $[-2, 2]$. You do not need to simplify your answers.

$f(-2) = 9(-2)^{\frac{4}{3}}e^{-\frac{2}{3}}$

$f(0) = 0$ ← min

$f(2) = 9(2)^{\frac{4}{3}}e^{\frac{2}{3}}$ ← max

B2 - The Mean Value Theorem (inspired by the class notes and past final worksheet problem 2)

(a) Find all values c guaranteed by the Mean Value Theorem when applied to the function $f(x) = \frac{x^3}{3}$ on the interval $[-\sqrt{3}, \sqrt{3}]$.

$$\frac{f(\sqrt{3}) - f(-\sqrt{3})}{\sqrt{3} - -\sqrt{3}} = \frac{\frac{(\sqrt{3})^3}{3} - \frac{(-\sqrt{3})^3}{3}}{2\sqrt{3}} = \frac{\sqrt{3} - -\sqrt{3}}{2\sqrt{3}} = 1$$

$$f'(x) = x^2 = 1 \rightarrow \boxed{c = \pm 1}$$

(b) Show that the function $f(x) = x^4 + e^x - x$ has exactly one point where the slope of the tangent line is 2. Be sure to fully justify your answer.

$$f'(x) = 4x^3 + e^x - 1 = 2 \rightarrow 4x^3 + e^x - 3 = 0$$

$g(x) = 4x^3 + e^x - 3$ is continuous and differentiable, so we can apply IVT and MVT.

$x=0$: $4 \cdot 0^3 + e^0 - 3 = -2 < 0$
$x=1$: $4 \cdot 1^3 + e^1 - 3 = 1 + e > 0$

$\Big\}$ There is some solution in $(0,1)$, by IVT.

$g'(x) = 12x^2 + e^x = 0$ has no solutions ($12x^2 + e^x$ is always positive)

so by MVT, $g(x) = 0$ has at most one solution.

At least one and at most one = exactly one

B3 - Curve sketching (inspired by past final worksheet problem 4 and the class notes)

Consider the following function along with its first derivative.

$$f(x) = \frac{x}{\sqrt{x^2+2x+1}} \qquad\qquad f'(x) = \frac{1}{(x+1)\sqrt{x^2+2x+1}}$$

(a) Find all horizontal asymptotes of $f(x)$.

$$\lim_{x\to\infty} \frac{x}{\sqrt{x^2+x+1}} = \lim_{x\to\infty} \frac{1}{\sqrt{1+\frac{1}{x}+\frac{1}{x^2}}} = 1 \longrightarrow \boxed{y=1}$$

$$\lim_{x\to-\infty} \frac{x}{\sqrt{x^2+x+1}} = \lim_{x\to\infty} \frac{-x}{\sqrt{x^2-x+1}} = \lim_{x\to\infty} \frac{-1}{\sqrt{1-\frac{1}{x}+\frac{1}{x^2}}} = -1 \longrightarrow \boxed{y=-1}$$

(b) Compute two one-sided limits at each vertical asymptote of $f(x)$.

$$f(x) = \frac{x}{\sqrt{(x+1)^2}}$$

$$\lim_{x\to-1^-} \frac{x}{\sqrt{(x+1)^2}} \approx \frac{-1}{\text{tiny}+\#} = -\infty$$

$$\lim_{x\to-1^+} \frac{x}{\sqrt{(x+1)^2}} \approx \frac{-1}{\text{tiny}+\#} = -\infty$$

(c) Find all intervals where $f(x)$ is decreasing.

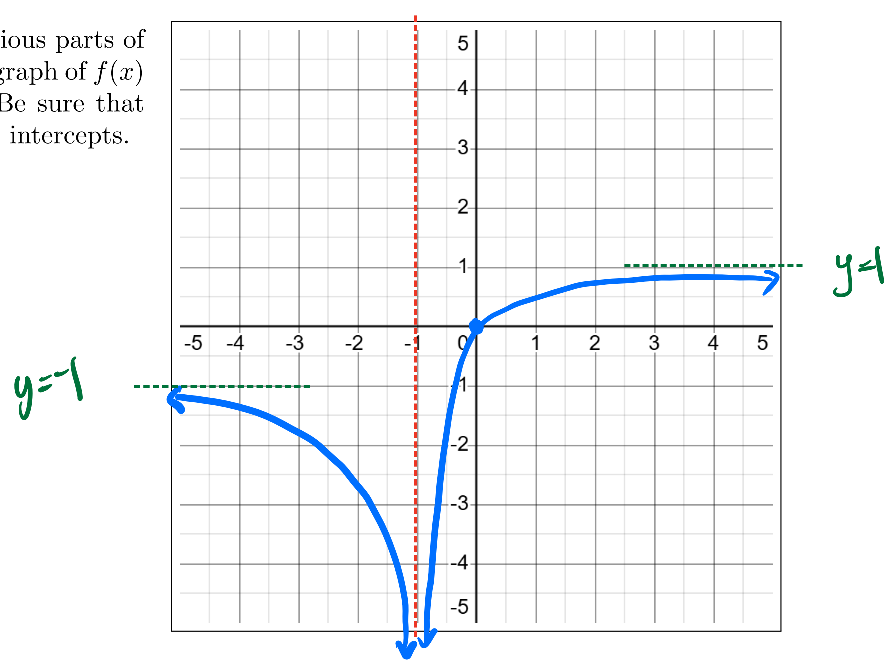
$f'(x)$ DNE @ $x=1$

$(-\infty,-1)$

$-$ $+$

-1

(d) Use the information from the previous parts of the problem to sketch an accurate graph of $f(x)$ on the axes shown to the right. Be sure that your sketch accurately reflects any intercepts.



B4 - Optimization (inspired by part 3 of the class notes and Brightspace problem 2)

(a) Consider the functions $f(x) = x^3$ and $g(x) = 27x$. What is the smallest vertical distance between these functions on the interval $[1, 5]$? Be sure to fully justify your answer.

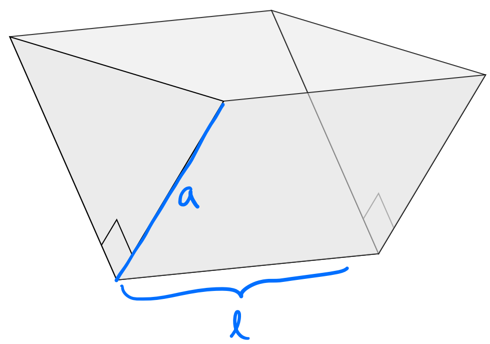
$D(x) = 27x - x^3 \qquad D'(x) = 27 - 3x^2 = 0 \rightarrow x = \pm 3$

$D(1) = 27 \cdot 1 - 1^3 = 26$

$D(3) = 27 \cdot 3 - 3^3 = 54$

$D(5) = 27 \cdot 5 - 5^3 = \boxed{10}$

(b) A trough with no top is built with two isosceles right triangles as ends, as shown on the right. If the volume of the trough must be 4 cubic meters, find the smallest area of material required to build it. The formula for the volume of the trough is the area of one of the triangles times the trough's length. Be sure to fully justify your answer and include units.



$\text{Area} = 2\ell a + 2 \cdot \frac{1}{2} a^2 = 2\ell a + a^2$

$\text{Volume} = \frac{1}{2} a^2 \cdot \ell = 4 \rightarrow \ell = \frac{8}{a^2}$

$A(a) = 2 \cdot \frac{8}{a^2} \cdot a + a^2 = \frac{16}{a} + a^2 \quad \text{on} \quad (0, \infty)$

$A'(a) = \frac{-16}{a^2} + 2a = 0 \rightarrow -16 + 2a^3 = 0 \rightarrow a = 2$

$\searrow \; - \quad\quad \nearrow \; +$

$0 \qquad 2$

$a=2$ is an absolute minimum (only critical point)

$A(2) = \frac{16}{2} + 2^2 = \boxed{12 \text{ m}^2}$

B5 - Antiderivatives (inspired by part 1 of the class notes and Brightspace problem 10)

(a) Suppose that $f(x)$ and $g(x)$ are two functions defined on an interval such that $f'(x) = g'(x)$. Suppose also that $f''(x)$ exists, and that $F(x)$ is an antiderivative of $f(x)$ and $G(x)$ is an antiderivative of $g(x)$. Which of the following must be true? Circle all that apply.

A. $f(a) - g(a) = f(b) - g(b)$ for any a and b in the interval.
B. $f(b) - f(a) = g(b) - g(a)$ for and a and b in the interval.
C. $f(a) = g(a)$ for any a in the interval.
D. $f''(a) = g''(a)$ for any a in the interval.
E. $F''(a) = G''(a)$ for any a in the interval.
F. $F(x) = G(x) + C$ for some constant C.
G. The graph of $f(x)$ is a vertical stretch of the graph of $g(x)$.
H. $a \cdot F(x) - G(x) + b$ is an antiderivative of $a \cdot f(x) - g(x) + b$ for any constants a and b.

(b) Suppose $h(x)$ is a function such that $h(1) = 2$, $h(-2) = -1$ and $h''(x) = 6$. Find two different antiderivatives of $h(x)$.

$h'(x) = 6x + A$

$h(x) = 3x^2 + Ax + B$

$h(1)=2$: $2 = 3 + A + B$

$h(-2)=-1$: $-\quad -1 = 12 - 2A + B$

$3 = -9 + 3A \to A = 4$

$\to 2 = 3 + 4 + B$

$\to B = -5$

$h(x) = 3x^2 + 4x - 5$

So any antiderivative has the form $x^3 + 2x^2 - 5x + C$, e.g.,

$\boxed{x^3 + 2x^2 - 5x + 1 \quad \text{and} \quad x^3 + 2x^2 - 5x - 1}$